Twisting elementary abelian groups via cocycles: from E_{16} to $D_8 \times C_2$

The non-abelian group $D_8 \times C_2$ can be constructed from the elementary abelian group $(C_2)^4$ by a single explicit 2-cocycle $\varphi(x,y) = x_1y_1 + x_2y_1$ valued in F_2, producing the twisted multiplication $(x_1,x_2,x_3,x_4) * (y_1,y_2,y_3,y_4) = (x_1+y_1, x_2+y_2, x_3+y_3, x_4+y_4+x_1y_1+x_2y_1)$ on the underlying set F_2^4. This construction is a central extension classified by the cohomology class $[x_1^2 + x_1x_2] \in H^2((C_2)^3, C_2)$, decomposing into a quadratic form (encoding element orders) and an alternating bilinear form (encoding commutators). The same algebraic mechanism — abelian vector space plus cocycle correction — independently governs the structure of the Pauli group in quantum error correction, the classification of extraspecial 2-groups via Arf invariants, and the relationship between Clifford algebras and Salingaros vee groups. What follows is a systematic treatment unifying these perspectives.

The cohomological classification machine

The foundational theorem, due to Eilenberg–MacLane (1942) and rooted in earlier work by Schur (1904, 1907) and Baer (1934), establishes a bijection between central extensions $1 \to A \to G \to Q \to 1$ with abelian kernel A and trivial Q-action, and the second cohomology group $H^2(Q, A)$. For our construction, $Q = (C_2)^3$ and $A = C_2$, so the relevant cohomology group is $\mathbf{H^2((C_2)^3, C_2) \cong (C_2)^6}$, a 6-dimensional F_2-vector space with **64 elements** corresponding to 64 distinct (though not all non-isomorphic) central extensions.

The structure of this cohomology group follows from the polynomial ring computation $H^*((C_2)^n, F_2) = F_2[x_1, \ldots, x_n]$, established rigorously in Adem–Milgram's *Cohomology of Finite Groups* (Springer, 2004, Theorem 4.1). The degree-2 part decomposes as

$$\mathbf{H^2((C_2)^n, C_2) = \text{span}\{x_i^2\} \oplus \text{span}\{x_ix_j : i < j\}}$$

with dimension $n + n(n-1)/2 = n(n+1)/2$. For $n = 3$ this gives $\mathbf{6 = 3 + 3}$. The universal coefficient theorem provides a complementary decomposition: the Ext^1 part (dimension n, corresponding to cup-squares x_i^2) classifies abelian extensions, while the $\text{Hom}(\Lambda^2 Q, A)$ part (dimension $n(n-1)/2 = 3$, corresponding to cup-products x_ix_j) encodes alternating bilinear forms — these capture all non-commutativity. Dipendra Prasad's "Notes on Central Extensions" (arXiv:1502.02140) gives an elegant treatment: $H^2(G, A) \cong \text{Sym}^2(G^\vee)$ for G abelian with trivial action over F_2, where $G^\vee = \text{Hom}(G, C_2)$. IIT Bombay

The Schur multiplier $M((C_2)^n) = H_2((C_2)^n, \mathbb{Z}) \cong (C_2)^{n(n-1)/2}$ — a classical result of Schur (1907), sharpened by Green's bound (1956) which shows equality holds if and only if the group is elementary abelian (Berkovich, *J. Algebra*, 1991). For $n = 3$, $M((C_2)^3) \cong (C_2)^3$; for $n = 4$, $M((C_2)^4) \cong (C_2)^6$. Brown's *Cohomology of Groups* (Springer GTM 87, 1982, Chapter IV) and Karpilovsky's *The Schur Multiplier* (LMS Monographs, 1987, Chapter 13) remain the standard references for these computations.

Which cocycle class builds $D_8 \times C_2$

Among the 64 elements of $H^2((C_2)^3, C_2)$, the action of $\mathrm{Aut}((C_2)^3) = GL(3, F_2)$ (order 168) partitions the extensions into orbits corresponding to distinct isomorphism classes. $D_8 \times C_2$ = SmallGroup(16, 11) sits in the central extension

$$\mathbf{1 \to \langle r^2 \rangle \cong C_2 \to D_8 \times C_2 \to (C_2)^3 \to 1}$$

where the quotient map sends the generators r, s, t of $D_8 \times C_2$ (with $r^4 = s^2 = t^2 = 1$, $srs = r^{-1}$, t central) to basis vectors of $(C_2)^3$. The center $Z(D_8 \times C_2) = \langle r^2, t \rangle \cong (C_2)^2$ and the commutator subgroup $[D_8 \times C_2, D_8 \times C_2] = \langle r^2 \rangle \cong C_2$ are confirmed by the Groupprops wiki entry for SmallGroup(16,11).

Choosing the section $\sigma(x_1, x_2, x_3) = r^{x_1} \cdot s^{x_2} \cdot t^{x_3}$, the cocycle φ defined by $\sigma(x) \cdot \sigma(y) = r^{2\varphi(x,y)} \cdot \sigma(x + y)$ evaluates to

$$\boldsymbol{\varphi(x, y) = x_1 y_1 + x_2 y_1 \pmod 2}$$

which is verified directly: $\sigma(1,0,0) \cdot \sigma(1,0,0) = r^2 = r^2 \cdot \sigma(0,0,0)$ gives $\varphi = 1$; $\sigma(0,1,0) \cdot \sigma(1,0,0) = s \cdot r = r^{-1}s = r^2 \cdot (rs) = r^2 \cdot \sigma(1,1,0)$ gives $\varphi = 1$; $\sigma(1,0,0) \cdot \sigma(0,1,0) = r \cdot s = rs = \sigma(1,1,0)$ gives $\varphi = 0$. The cocycle condition $\varphi(x,y) + \varphi(x+y,z) = \varphi(x,y+z) + \varphi(y,z)$ can be checked algebraically. This cocycle represents the cohomology class $\mathbf{[\varphi] = x_1^2 + x_1 x_2 \in H^2((C_2)^3, F_2)}$, combining a cup-square (x_1^2: the symmetric/quadratic contribution encoding the fact that r has order 4) and a cup-product ($x_1 x_2$: the alternating contribution encoding the commutator $[r,s] = r^2$).

The full group operation on F_2^4 that recovers $D_8 \times C_2$ is therefore:

$$\mathbf{(x_1,x_2,x_3,x_4) * (y_1,y_2,y_3,y_4) = (x_1+y_1, x_2+y_2, x_3+y_3, x_4+y_4+x_1y_1+x_2y_1)}$$

where x_4 is the "kernel coordinate" corresponding to the central subgroup $\langle r^2 \rangle$. This formula is self-contained: starting from pure bitwise XOR on F_2^4, the single correction term $x_1y_1 + x_2y_1$ in the fourth coordinate introduces all the non-commutativity and non-trivial element orders of $D_8 \times C_2$. Verification: (1,0,0,0) has order 4 since $(1,0,0,0)^2 = (0,0,0,1)$ and $(0,0,0,1)^2 = (0,0,0,0)$; the commutator $[(1,0,0,0), (0,1,0,0)] = (0,0,0,1) = r^2$; and (0,0,1,0) commutes with everything since x_3 does not appear in φ.

For comparison, $H^2(V_4, C_2) \cong (C_2)^3$ classifies extensions of $V_4 = (C_2)^2$ by C_2 — the Groupprops wiki page on this group enumerates all eight cocycle classes and their extensions: the trivial class gives $(C_2)^3$; the classes x_1^2 and x_2^2 each give $C_4 \times C_2$; the class x_1x_2 gives D_8; and the class $x_1^2 + x_2^2 + x_1x_2$ gives Q_8. The $D_8 \times C_2$ cocycle on $(C_2)^3$ is precisely the "D_8 cocycle x_1x_2 on $(C_2)^2$" extended trivially by one additional coordinate.

Quadratic forms, the Arf invariant, and extraspecial groups

The cocycle φ decomposes canonically into two components encoding distinct group-theoretic data. The **alternating part** $\beta(x,y) = \varphi(x,y) + \varphi(y,x) = x_1y_2 + x_2y_1$ is a symplectic form of rank 2 on F_2^3 with one-dimensional radical spanned by e_3. It captures the commutator: $[\sigma(x), \sigma(y)] = r^{2\beta(x,y)}$. The **quadratic part** $q(x) = \varphi(x,x) = x_1$ (since $x_1^2 = x_1$ over F_2) captures the squaring map: $\sigma(x)^2 = r^{2q(x)}$. Elements with $q = 0$ lift to involutions; elements with $q = 1$ lift to order-4 elements. This decomposition into (quadratic form, alternating bilinear form) is precisely the

structure Quillen exploited in his foundational paper "The mod 2 cohomology rings of extra-special 2-groups and the spinor groups" (*Math. Ann.* 194, 1971, 197–212).

Quillen classified the cohomological behavior of extraspecial 2-groups of order **2^{1+2n}** (Wikipedia) using the quadratic form $Q: V \to F_2$ on the central quotient $V = G/Z(G)$, where $Q(gZ) = 0$ if $g^2 = 1$ and $Q(gZ) = 1$ if $g^2 = z$ (the central generator). (Wikipedia) The polarization $B(x,y) = Q(x+y) + Q(x) + Q(y)$ is a non-degenerate alternating (symplectic) form on $V \cong F_2^{2n}$. (Springer) Quillen identified three cases: **"real"** when $\dim V_1 = 0$ and $\mathrm{Arf}(Q) = 0$ (Cocalc) (central products of D_8's), **"quaternion"** when $\dim V_1 = 0$ and $\mathrm{Arf}(Q) = 1$ (odd number of Q_8 factors), and **"complex"** when $V_1 \neq 0$ (Q is singular, i.e., has nontrivial radical). (Cocalc) The Arf invariant — originally defined by Arf (1941) for nonsingular quadratic forms over perfect fields of characteristic 2, with antecedents in Dickson (1901) — equals $\Sigma\, Q(e_i)Q(f_i) \in F_2$ for any symplectic basis $\{e_i, f_i\}$. It distinguishes the two isomorphism types of extraspecial groups at each order: (Wikipedia) **Arf = 0 gives the "plus type" 2_+^{1+2n}** (Wikipedia) (central product of D_8's) and **Arf = 1 gives the "minus type" 2_-^{1+2n}** (odd number of Q_8 factors). (Wikipedia)

$D_8 \times C_2$ is not itself extraspecial — its center $(C_2)^2$ has order 4, not prime — but it belongs to Quillen's "complex case." The quadratic form $q(x_1,x_2,x_3) = x_1$ on F_2^3 has radical $\ker(B) = \langle e_3 \rangle$ of dimension 1; restricting to the non-degenerate quotient F_2^2 gives a form with Arf invariant 0. $D_8 \times C_2$ is thus an **"almost extraspecial"** group: D_8 extended by a direct C_2 factor that lies in the center but outside the commutator subgroup. Gorenstein's *Finite Groups* (Harper & Row, 1968, Chapter 5), Griess's paper "Automorphisms of extra special groups and nonvanishing degree 2 cohomology" (*Pacific J. Math.* 48, 1973, 403–422), and Craven's lecture notes "The Theory of p-Groups" (University of Birmingham) give detailed treatments of this classification.

The Pauli group and quantum phase correction

The identical algebraic mechanism governs the n-qubit Pauli group in quantum error correction. The Pauli group P_n modulo scalar phases acts on the Hilbert space of n qubits with elements labeled by binary vectors $(a|b) \in F_2^{2n}$. (University of California, Be...) (HandWiki) The quotient $P_n/Z(P_n) \cong F_2^{2n}$ is an elementary abelian 2-group, but the actual group multiplication carries a cocycle correction:

$$\mathbf{E(a,b)\cdot E(a',b') = (-1)^{c(v,w)}\, E(a+a',\, b+b')}$$

where $c: F_2^{2n} \times F_2^{2n} \to F_2$ is a 2-cocycle. This formula appears explicitly in Rengaswamy, Calderbank, Kadhe, and Pfister, "Synthesis of Logical Clifford Operators via Symplectic Geometry" (*IEEE Trans. Inform. Theory*, 2020; arXiv:1803.06987), using the specific form $D(a,b)\cdot D(a',b') = (-1)^{a'\cdot b^T}\, D(a+a',\, b+b')$. (arXiv) The **alternating part** of this cocycle is the symplectic inner product $\langle v,w \rangle_s = a\cdot b' + a'\cdot b \pmod 2$, which captures commutativity: two Pauli operators commute if and only if their symplectic inner product vanishes. (University of California, Be...) This is the content of the Calderbank–Rains–Shor–Sloane (CRSS) framework ("Quantum Error Correction via Codes over GF(4)," *IEEE Trans. Inform. Theory* 44(4), 1998; arXiv:quant-ph/9608006), which translates quantum error correction into the language of self-orthogonal codes under the symplectic inner product. (arXiv)

For a **single qubit** (n = 1), the Pauli group modulo ±i has order 8 and is isomorphic to D_8. The Wikipedia article on the Heisenberg group confirms: "The Heisenberg group modulo 2 is of order 8 and is isomorphic to the dihedral group D_4" (in the convention where D_4 has order 8). (Wikipedia) The central extension is $1 \to \{\pm I\} \to D_8 \to F_2^2 \to 1$, classified by the symplectic form $\omega(v,w) = a_1b_2 + a_2b_1$ on F_2^2. The cocycle $c((a_1,a_2),(b_1,b_2)) = a_2b_1$ satisfies $c(v,w) + c(w,v) = \omega(v,w)$. This is identical in structure to the D_8 portion of the $D_8 \times C_2$ cocycle.

The group $D_8 \times C_2$ therefore corresponds to the single-qubit Pauli group (mod ±i) augmented by one "inert" classical bit. The cocycle $\varphi(x,y) = x_1y_1 + x_2y_1$ on F_2^3 is precisely the single-qubit phase correction on the first two coordinates, extended trivially to a third. In stabilizer code language, this third coordinate sits in the radical of the symplectic form — it generates a one-dimensional totally isotropic subspace that commutes with all operators. (HandWiki) Gottesman's thesis ("Stabilizer Codes and Quantum Error Correction," Caltech, 1997; arXiv:quant-ph/9705052) (ScienceDirect) (arXiv) and (arXiv) Knill's "Group Representations, Error Bases and Quantum Codes" (1996; arXiv:quant-ph/9608049) (arXiv) established the general framework: nice error bases are projective representations of finite groups (Springer) where the projective factor ω_{g,h} in E_g · E_h = ω_{g,h} E_{gh} is exactly the 2-cocycle of the central extension. Appleby's work on SIC-POVMs (*J. Math. Phys.* 46, 052107, 2005; arXiv:quant-ph/0412001) (arXiv) (ResearchGate) and (OSTI) Gross's "Hudson's theorem for finite-dimensional quantum systems" (*J. Math. Phys.* 47, 122107, 2006; arXiv:quant-ph/0602001) (arXiv) further develop the discrete phase-space perspective, using the Weyl commutation relation W_x W_y = τ^{[x,y]} W_{x+y}.

Dehaene and De Moor ("Clifford group, stabilizer states, and linear and quadratic operations over GF(2)," *Phys. Rev. A* 68, 042318, 2003; arXiv:quant-ph/0304125) made the connection to quadratic forms fully explicit: Clifford group operations correspond to symplectic matrices acting on the binary phase space, and the phase tracking under conjugation involves quadratic forms over F_2 (arXiv) — precisely the algebraic data classifying our central extensions.

Clifford algebras, Salingaros vee groups, and combinatorial designs

The cocycle-twist construction has a third incarnation in the theory of Clifford algebras over finite fields. Given an n-dimensional F_2-vector space V with quadratic form Q, the Clifford algebra Cl(V, Q) is isomorphic to a **twisted group algebra** $F_t[(C_2)^n]$ where the twist encodes Q. Albuquerque and Majid established this identification, and it was systematically developed by Salingaros in a series of papers (*J. Math. Phys.* 22, 1981; 23, 1982; 25, 1984) classifying the "vee groups" G_{p,q} ⊂ Cl_{p,q}. These vee groups of order 2^{n+1} fall into five classes:

- **N_{2k-1}**: central product of k copies of D_8 (extraspecial, Arf = 0)
- **N_{2k}**: odd number of Q_8 central factors (extraspecial, Arf = 1) (ResearchGate)
- **Ω_{2k-1}**: $N_{2k-1} \circ (C_2 \times C_2)$ (Tennessee Tech University) **— $D_8 \times C_2$ is the case k = 1**
- **Ω_{2k}**: $N_{2k} \circ (C_2 \times C_2)$
- **S_k**: central products with C_4

Abłamowicz, Varahagiri, and Walley (*Adv. Appl. Clifford Algebras* 28, 2018) proved that every Clifford algebra Cl_{p,q} is R-isomorphic to R[G_{p,q}]/(1+τ) where τ is a central involution of the

vee group. (Springer) The key insight: **Clifford algebras are quotients of twisted group algebras of elementary abelian 2-groups, and the twist is precisely the 2-cocycle we have been studying.** For $D_8 \times C_2 = \Omega_1$, the corresponding Clifford algebra encodes a 3-dimensional quadratic space with a 1-dimensional radical. Quillen's computation of $H^*(BG, F_2)$ for extraspecial G uses precisely the ideal generated by Q and its Steenrod operations. (ResearchGate)

In algebraic combinatorics, the cocycle-to-group construction connects to several design-theoretic objects. Horadam (*Des. Codes Cryptogr.*) proved that semiregular central relative difference sets in a group E avoiding a central subgroup C are equivalent to **orthogonal cocycles**. (Springer) Presemifield multiplication defines "LP cocycles" (linearized permutation cocycles) on elementary abelian groups, (Springer) linking semifield theory (Knuth, *J. Algebra* 2, 1965; Kantor, *J. Algebra* 270, 2003; Lavrauw–Polverino survey, 2010) to the cocycle framework. Kantor and Williams ("Symplectic semifield planes and $\mathbb{Z}_4$-linear codes," *Trans. AMS* 356, 2004) demonstrated how characteristic-2 semifields, orthogonal spreads, and $\mathbb{Z}_4$-linear Kerdock/Preparata codes arise from the same (Semantic Scholar) "abelian space + cocycle" mechanism. The quadratic form that classifies the extension simultaneously determines the combinatorial properties of associated codes and designs. Wood's paper "Witt's extension theorem for mod four valued quadratic forms" (*Trans. AMS*, 1993) extends the Arf invariant classification to $\mathbb{Z}/4$-valued quadratic forms, connecting to doubly-even self-orthogonal codes interpreted as totally isotropic subspaces (University of Edinburgh) — exactly the stabilizer codes of quantum information.

Pakianathan and Yalçın ("Quadratic Maps and Bockstein Closed Group Extensions," arXiv:math/0606374, *J. Algebra*, 2012) provided the most comprehensive modern treatment: central extensions of elementary abelian 2-groups by elementary abelian 2-groups are determined uniquely by quadratic maps, and the "Bockstein closed" condition characterizes extensions whose mod-2 cohomology rings have particularly clean structure. (arXiv)

The unified structure theorem

The three perspectives — group cohomology, quantum information, Clifford algebras — describe the same mathematical object. A formal summary:

Theorem (synthesis). Let $V = F_2^n$ be an elementary abelian 2-group, $K \cong C_2$ a designated coordinate subgroup, and $\varphi: V/K \times V/K \to K$ a normalized 2-cocycle. The twisted operation $x * y = x + y + \varphi(\pi(x), \pi(y))$ on V (where $\pi: V \to V/K$ is the projection) yields a group G fitting in the central extension $1 \to K \to G \to V/K \to 1$. The cocycle φ decomposes into:

- A **quadratic form** $q: V/K \to F_2$ defined by $q(x) = \varphi(x,x)$, encoding the squaring map (element orders in G).
- An **alternating bilinear form** $B(x,y) = \varphi(x,y) + \varphi(y,x)$, encoding the commutator map $[\cdot,\cdot]: G \times G \to K$.
- If B is non-degenerate on a 2m-dimensional subspace $W \subset V/K$ with complement $W^\perp$ (the radical), then $G \cong E * (C_2)^{\dim W^\perp}$ where E is an extraspecial 2-group of order 2^{1+2m} whose type (plus or minus) is determined by the Arf invariant of $q|_W$.

For the specific case $V = F_2^4$, $K = \langle e_4 \rangle$, and $\varphi(x,y) = x_1y_1 + x_2y_1$: the bilinear form B has rank 2 on span$\{e_1, e_2\}$ with radical $\langle e_3 \rangle$; the quadratic form $q(x) = x_1$ restricts to a form with Arf invariant 0 on the non-degenerate part; and the resulting group is $\mathbf{D_8 \times C_2}$ (Arf = 0, one radical direction). Changing the cocycle to $x_1^2 + x_2^2 + x_1x_2$ (Arf = 1, no radical) would yield Q_8 instead.

In quantum information, V/K = F_2^{2n} is the binary phase space of n qubits, B is the symplectic inner product governing Pauli commutation, q encodes the phase tracking under operator multiplication, and the Clifford group Cliff_n acts as $\mathbf{Sp(2n, F_2)}$ — the automorphism group preserving B. (arXiv +2) In Clifford algebra theory, V/K carries the quadratic form of signature (p,q) with p + q = n, and the (Springer) vee group G is the Salingaros group whose class is determined by (rank B, Arf q). In combinatorics, orthogonal cocycles on V/K produce relative difference sets, (Springer) and the resulting designs inherit their combinatorial parameters from the quadratic/symplectic data.

Conclusion

The construction of $D_8 \times C_2$ from $(C_2)^4$ via a 2-cocycle is far more than a pedagogical exercise in group extensions. It is a minimal non-trivial instance of a mechanism that recurs across mathematics: **non-commutativity on a binary vector space is always governed by a quadratic form and its associated symplectic structure, mediated by a 2-cocycle valued in a central C_2 subgroup.** The cohomology class $[x_1^2 + x_1x_2] \in H^2((C_2)^3, C_2)$ that builds $D_8 \times C_2$ is the exact algebraic analogue of the phase correction $(-1)^{a' \cdot b^T}$ in the single-qubit Pauli multiplication formula — both are instances of the same universal construction classified by $H^2(V, K)$. The Arf invariant provides the finest discrete invariant distinguishing extension types (D_8 vs. Q_8), (Wikipedia) while the radical of the bilinear form accounts for "inert" direct factors (the $\times C_2$ in $D_8 \times C_2$). This triple perspective — Eilenberg–MacLane cohomology, Gottesman–CRSS symplectic formalism, and Salingaros–Quillen Clifford classification — constitutes a single coherent theory of cocycle-twisted operations on elementary abelian 2-groups, with the quadratic form over F_2 as its central organizing object.